FLOAT: Flood-Aware Trajectories for Modeling the Time-Dependent Evacuation Routing with Risk Consideration

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***Abstract*—In flood-prone urban environments, informed and adaptive routing is essential for ensuring effective evacuation. This study presents a flood-aware routing framework for dynamic evacuation planning, formulated as a time-dependent Arc Routing Problem (ARP) with edge costs modulated by tunable parameters: λ to balance travel time and flood risk, α as a flood-speed penalty factor, and β as a one-way traversal penalty to flexibly override directionality in extreme conditions. We evaluate the algorithm from multiple perspectives: (1) performance metrics such as travel time, cumulative flood depth, flood risk, and flooded segment count; (2) spatial behavior and network connectivity implications via route usage distribution and spatial autocorrelation; and (3) sensitivity to key parameters under varying levels of risk aversion and flood avoidance. Using the road network of Marikina City, Philippines comprised of 4892 edges, 3309 nodes, and simulated flood depths at one-minute intervals over one hour, we simulate 20,000 evacuation scenarios comparing the Flood-Aware Routing (FAR) against a Shortest Path Routing (SPR) baseline. Results demonstrate that the FAR approach effectively reduces flood exposure and traverses fewer inundated segments, with trade-offs in travel time. Sensitivity analysis indicates that higher values of λ and α enhance flood avoidance but can lead to longer and occasionally suboptimal routes due to the algorithm's local and greedy decision-making structure. In certain instances, the FAR algorithm is observed to favor short-term avoidance over longer-term route efficiency, highlighting a need for improved global reasoning. Moran's I analysis reveals spatial preference for low-risk corridors, indicating that certain road segments consistently emerge as favorable pathways under dynamic flood conditions. Moreover, the observed spatial clustering affirms that flood-aware routing does not merely result in random path choices, but rather forms coherent spatial patterns shaped by underlying flood level information, topography, and network connectivity.**

***Index Terms*—time-dependent routing, arc routing problem, dynamic routing, flood routing, risk aware routing**

I. Introduction

Routing is fundamental to a wide range of operational problems including the snow-plow routing of winter maintenance fleets [1], scheduled parcel delivery [2], resource allocation [3], and urban waste collection [4]. While static routing methods can perform adequately under stable, predictable conditions, they are ill-suited to time-sensitive scenarios such as evacuations, fire responses, and flooding where network usability can change rapidly. In evacuation planning, conventional approaches assume fixed road conditions, ignoring dynamic hazards like rising floodwaters. This assumption can leave previously viable routes suddenly impassable. To overcome this, several studies have proposed dynamic evacuation frameworks that integrate real-time flood data and continuously re-optimize routes [5]. For instance, [6] incorporate spatiotemporal inundation forecasts into route-cost estimations, and [7] leverage a cellular-automata model, factoring in human response time, topography, and hydrodynamic conditions. They demonstrate significant performance gains, reducing trapped individuals by 32.7%, average path lengths by 34.0%, and maximum path lengths by 7.9% compared to static A-star routing. Furthermore, [8] enhanced Dijkstra's algorithm by incorporating three-dimensional conditions, flood-risk weighting and integration with shelter locations. Time-aware evacuation modeling has shown practical relevance when applied to complex, real-world urban environments. In the work of [9], pedestrian behavior during flood events was explored using microsimulation, revealing that walking dynamics are highly sensitive to flood-related variables such as water depth and flow conditions. Complementarily, [10]

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examined the implementation of dynamic routing in urban evacuations, finding that the effectiveness of routing strategies can differ significantly across locations, which emphasizes the importance of designing evacuation plans that are responsive to spatial variability. Dynamic route optimization is not confined to flood hazards. For example, [11] developed a safe-driving routing framework that estimates crash probabilities based on traffic levels, road design, and environmental conditions. It is has also been applied to adapt to customer requests arriving in real time under traffic flow uncertainty in logistics scenarios [12], [13]. These approaches share a common theme: travel cost and risk are updated in response to changing environmental conditions. A specific class of such methods focuses on dynamic edge-cost modification, where each road segment's weight changes in response to new data. Like the Risk-Aware Graph Search (RAGS) of [14], which treats edge costs as probability distributions rather than fixed values such that during traversal, actual costs are revealed and used to balance exploration of uncertain edges against exploitation of low-risk options, thus avoiding high-cost scenarios. While the transportation resilience field has extensively embraced risk-based methodologies to strengthen disaster response and infrastructure robustness, significant technical shortcomings remain. Specifically, there is a lack of fine-grained, temporal modeling of flood dynamics, especially in contexts like the Philippines. Prior studies in such settings have not analyzed inundation patterns and its implications on road network vulnerability, travel time, and flood exposure. Moreover, existing dynamic-routing methods rarely combine temporal segmentation of flood travel times with synchronous risk assessment. And even among those that do, there is little attention to conducting sensitivity analyses investigating how parameter variations affect the balance between evacuation speed and safety in actual urban landscapes. To address this, the study aims to: (1) Develop a Time-Dependent Routing Framework that captures both spatial and temporal variability by incorporating travel distance and flood-induced risk into an edge cost function, (2) Apply the Model to a real-world urban network of Marikina City, Metro Manila, Philippines, (3) Compare Flood-Aware and Baseline Routing Strategies through key performance metrics, spatial distribution of network usage, as well as identifying critical links under varying hazard conditions, (4) Conduct Sensitivity Analysis on Key Model Parameters to understand how variations in essential model parameters—such as the flood threshold, slowdown factor, and risk-weight coefficient affect routing outcomes.

II. Methodology

A. Problem Definition

The evacuation routing problem is formulated as an Arc Routing Problem (ARP). The network is modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the set of vertices $\mathcal{V}$ represents road intersections and the set of edges $\mathcal{E}$ represents street segments connecting these nodes. Each edge $e \in \mathcal{E}$ is associated with the following attributes: spatial geometry, elevation, flood level time series, road length, maximum allowed speed, and street type. It is assumed that along these street segments, evacuees are located; however, the amount of such in this study is considered negligible. The objective is to construct a route that begins and ends at the depot v_0, and visits the given required edge at least once, while minimizing the total cost of the route under the effect of time-varying and spatially heterogeneous flood water level conditions. Unlike self-evacuation models where individuals independently determine their optimal paths, this formulation assumes vehicle-based evacuation and abstracts away drop-off points along the path. The model also assumes single evacuation vehicle dispatched from a depot node $v_0 \in \mathcal{V}$ and unbounded capacity such that there is no requirement to perform intermediate unloading during service. The subset of edges that must be visited is denoted by $\mathcal{R} \subseteq \mathcal{E}$, referred to as required edges, and traversal of an edge in $\mathcal{R}$, in either direction is considered sufficient. This assumes that such move is enough to evacuate the individuals located along that required edge, and after such the need to return to depot, the starting point, will be trigerred to finish the closed path. The model incorporates time-dependence by computing an edge cost as a function of the current flood-level induced by currently in time t. Two considerations are introduced to handle edge cases. First, if there are multiple neighboring edges from a node having identical costs at a given time, a randomized tie-breaking strategy is employed to avoid deterministic bias and increase route exploration. Second, the model allows for the inclusion of one-way streets that would otherwise be prohibited under normal traffic regulations. Instead of strictly excluding these segments, the model applies a scalar penalty to the cost of traversing such edge in the wrong direction. This approach ensures that during extreme flooding scenarios, directionality constraints can be conditionally relaxed when the trade-off in terms of risk reduction or travel time improvement is substantial.

B. Flood and Risk Modelling

The framework is designed to quantify edge-level flood risk dynamically such that when a vehicle traverses a road network, discrete-time flood predictions and continuous vehicle movement are tallied. For each edge $(u, v) \in \mathcal{E}$, we define a *flood level time series* $\mathbf{F}^{u,v}$, which is a vector of length $N = T/\Delta t$. The ith component $f_i^{u,v}$ of this vector represents the average water depth on segment (u, v) during time window $[(i-1)\Delta t, i\Delta t)$. Formally, we define it as $\mathbf{F}^{u,v} = [f_1^{u,v}, f_2^{u,v}, \ldots, f_N^{u,v}], \quad \forall (u, v) \in \mathcal{E}$. Flooded conditions occur when a threshold is reached, and under such conditions, both risk and effective speed on the edge are altered. A time-dependent risk model is proposed for vehicle navigation in urban environments under flooding conditions. We define a global continuous time variable $t \in \mathbb{R}_{\geq 0}$ that serves as the reference clock for vehicle movement. Flood levels, however, are updated in discrete intervals, with update times denoted by $\{t_k\}_{k=0}^{N}$ where $t_{k+1} - t_k = \Delta t$ is fixed, in this case, assumed to be changing every minute. Let $G = (V, E)$ be the road network, where each edge $(u, v) \in E$ corresponds to

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a street segment with associated attributes such as length $l_{(u,v)}$ and maximum speed $s_{(u,v)}$. Each vehicle follows deterministic travel times between nodes, and decision-making is assumed to be instantaneous and not to consume traversal time. We define two types of edge-level travel time:

- **Canonical time** $\tau^{(u,v)}(t)$: the nominal travel time over edge (u,v) at time t, given by $\tau^{(u,v)}(t) = \frac{l_{(u,v)}}{s_{(u,v)}}$
- **Effective time** $\tau'^{(u,v)}(t)$: the actual traversal time accounting for flood impact, defined using the adjusted speed $s'_{(u,v)}(t)$ as: $\tau'^{(u,v)}(t) = \frac{l_{(u,v)}}{s'_{(u,v)}(t)}$

The adjusted speed is defined based on the flood level $f_{(u,v)}(t)$ at time t and a global flood threshold ϱ:

$$s'_{(u,v)}(t) = \begin{cases} s_{(u,v)}, & \text{if } f_{(u,v)}(t) \le \varrho \quad \text{(safe condition)} \\ \alpha \cdot s_{(u,v)}, & \text{if } f_{(u,v)}(t) > \varrho \quad \text{(flooded condition)} \end{cases} \tag{1}$$

where $\alpha \in (0,1)$ is a slowdown factor under flooding. Let a vehicle start traversal of edge (u,v) at global time t_0. The canonical traversal ends at $t_1 = t_0 + \tau^{(u,v)}(t_0)$. Since flood levels are updated at discrete times, the interval $[t_0, t_1]$ may span multiple flood update points. We define a composite **risk function** for traversing edge (u,v) at time t:

$$r^{(u,v)}(t) = \phi\left(f_t^{(u,v)}\right) \cdot \psi\left(h_{(u,v)}, f_t^{(u,v)}\right) \cdot \eta\left(\tau'^{(u,v)}(t)\right) \tag{2}$$

where $\phi(f) = \min\left(\frac{f}{f_{\max}}, 1\right)$ captures flood depth sensitivity, with $f_{\max} = 1.5$ meters, $\psi(h,f) = 1 - \tanh\left(\frac{h-10f}{100}\right)$ adjusts risk based on elevation resistance to flood, $\eta(\tau) = \min\left(\frac{\tau}{\tau_{\max}}, 1\right)$ represents exposure time, with $\tau_{\max} = 180$ seconds. The total traversal cost combines travel time and risk:

$$c^{(u,v)}(t) = \lambda \cdot \tau'^{(u,v)}(t) + \alpha \cdot r^{(u,v)}(t) + \beta \cdot \delta^{(u,v)} \tag{3}$$

where $\lambda \in [0,1]$ is a tunable parameter balancing speed and safety, β is the one-way penalty weight and $\delta^{u,v}$ is a binary variable specifying if the edge is oneway or not. The effective travel time $\tau'^{(u,v)}(t)$ may vary depending on the flooding status of each subsegment, especially if the vehicle enters and exits different flood regimes within a single edge. We define further $v_0 \in \mathcal{V}$ be the depot and $(u^*, v^*) \in \mathcal{R}$ be the required edge. A feasible round-trip path $\mathcal{P}$ must start and end at v_0 while visiting (u^*, v^*). For any candidate path $\mathcal{P} = [v_0, v_1, \ldots, v_k, v_0]$, the cumulative travel time and risk are computed as: $\tau'(\mathcal{P}, t_0) = \sum_{i=0}^{k-1} \tau'^{(v_i,v_{i+1})}(t_i)$ and $r(\mathcal{P}, t_0) = \sum_{i=0}^{k-1} r^{(v_i,v_{i+1})}(t_i)$ where t_i is the start time of the i-th edge traversal, updated cumulatively as:

$$t_{i+1} = t_i + \tau'^{(v_i,v_{i+1})}(t_i) \tag{4}$$

The effective travel time $\tau'^{(u,v)}(t)$ may vary depending on the flooding status of each subsegment, especially if the vehicle enters and exits different flood regimes within a single edge. From this framework, this model can be integrated with routing algorithms that optimize for minimal risk or balanced trade-offs between risk and travel time.

C. Illustration of Road Selection Cases

To demonstrate the proposed flood-aware routing framework, we consider a simplified schematic of a road network as shown in Figure 1, where a vehicle at node A must choose among three candidate streets to continue its journey. The relevant characteristics of each street, including length, flood level, effective speed, travel time, and computed risk, are summarized in Table I. Assume a canonical speed of 10 m/s on all streets under dry conditions, a flood threshold $\varrho = 0.5$ m, a slowdown factor $\alpha = 0.5$, and current time $t = 5$ seconds, which lies within the first flood time window $[0, 60)$.

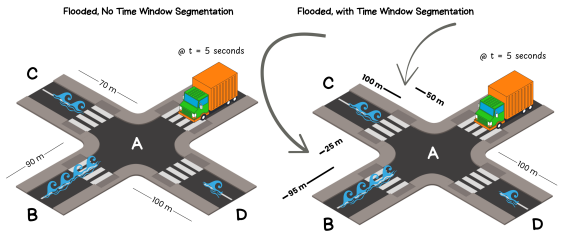


Fig. 1. Illustrative routing scenario for a vehicle departing from node A.

TABLE I
ROUTING METRICS FOR CANDIDATE STREETS FROM NODE A

Edge	Length (m)	Flood (m)	Speed (m/s)	Time (s)	Risk
(A,C)	70	0.9	5	14	14
(A,B)	90	0.5	10	9	0
(A,D)	100	0.1	10	10	0

The vehicle's route selection is governed by a dynamic cost function that jointly considers travel time and flood-induced risk: $c_{(u,v)}(t) = \lambda \cdot \tau'_{(u,v)}(t) + (1-\lambda) \cdot r^{(u,v)}(t)$, where $\tau'_{(u,v)}(t)$ is the effective travel time and $r^{(u,v)}(t)$ is the cumulative flood risk for edge (u,v) at time t. The parameter $\lambda \in [0,1]$ balances speed versus safety. Assuming a balanced trade-off with $\lambda = 0.5$, the costs for the three edges are computed as follows: (1) $c_{(A,C)} = 0.5 \cdot 14 + 0.5 \cdot 14 = 14$, (2) $c_{(A,B)} = 0.5 \cdot 9 + 0.5 \cdot 0 = 4.5$, (3) $c_{(A,D)} = 0.5 \cdot 10 + 0.5 \cdot 0 = 5$. Among the three candidates, edge (A,B) presents the minimum cost and is therefore selected as the optimal path under current flood conditions. Notably, edge (A,D) is also viable, offering zero risk at the expense of one additional second of travel time. If an edge traversal crosses into a new flood evaluation window, the time-dependent analysis must segment the trip accordingly. Let traversal begin at t_0 and conclude at $t_1 = t_0 + \tau'_{(u,v)}(t_0)$. If t_k marks a flood window boundary with $t_0 < t_k < t_1$, the travel must be partitioned into sub-intervals over $[t_0, t_k)$ and $[t_k, t_1)$. To further demonstrate, we revisit the schematic road network with the vehicle positioned at node A. The vehicle again evaluates three outgoing paths: (A,B), (A,C), and (A,D). In this scenario, we assume the length of each edge is sufficient to span across two flood update windows. Flood updates occur every $\Delta t = 60$ seconds, and the current global time is $t = 55$ seconds—just before the transition

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into the second window $[60, 120)$. Global parameters remain consistent with earlier assumptions: (1) Canonical speed: 10 m/s, (2) Flood threshold: $\varrho = 0.5$ m, and (3) Slowdown factor: $\alpha = 0.5$

TABLE II
FLOOD PROBABILITIES ACROSS TIME SEGMENTS

Edge	**Flood in** $[0, 60)$	**Flood in** $[60, 120)$
(A, B)	0.6	0.3
(A, C)	0.4	0.7
(A, D)	0.1	0.1

The time-dependent flood levels across windows are specified for each edge as seen in Table II. Edge lengths are given as: 120 m for (A, B), 150 m for (A, C), and 100 m for (A, D). The traversal time and associated risk are computed via segmentation aligned with flood time windows. During the first 5 seconds of Edge (A, B), $(t \in [55, 60))$, flood level $f = 0.6$ exceeds the threshold, so the vehicle proceeds at $s_1 = 0.5 \times 10 = 5$ m/s and covers $5 \times 5 = 25$ m. The remaining 95 m is traveled in the second window where $f = 0.3 < 0.5$, hence at full speed $s_2 = 10$ m/s. The corresponding time is $95/10 = 9.5$ s, yielding a total traversal time of 14.5 s. The risk accumulated is $r_1 = 1 \cdot 5 = 5$ in window one and $r_2 = 0$ in window two, totaling 5 units. For Edge (A, C), the flood level in the first window is $f = 0.4 < 0.5$, so the vehicle travels at 10 m/s, covering $5 \times 10 = 50$ m. The remaining 100 m is traversed during window two where flood level rises to $0.7 > 0.5$, reducing speed to $s_2 = 5$ m/s. The duration is $100/5 = 20$ s. Total travel time is thus 25 s, with $r_1 = 0$ and $r_2 = 20$, resulting in a cumulative risk of 20. For Edge (A, D), the flood level is consistently below the threshold in both windows. The entire segment is traversed at 10 m/s, yielding a total time of $100/10 = 10$ s, and zero risk. Based on the computed dynamic costs shown in Table III, the optimal choice is edge (A, D), which offers both the shortest travel time and zero risk exposure. Although edge (A, B) shows a moderate cost due to recovery of speed in the second window, it is still suboptimal relative to the dry edge. Edge (A, C) incurs the highest cost due to delayed traversal through a high-risk segment in the second time window.

TABLE III
TEMPORAL WINDOW ROUTING SUMMARY

Edge	**Travel Time (s)**	**Risk**	**Cost** $(\lambda = 0.5)$
(A, B)	14.5	5	$0.5 \cdot 14.5 + 0.5 \cdot 5 = 9.75$
(A, C)	25	20	$0.5 \cdot 25 + 0.5 \cdot 20 = 22.5$
(A, D)	10	0	$0.5 \cdot 10 + 0.5 \cdot 0 = 5$

III. EXPERIMENTS

We evaluate our proposed flood-aware routing algorithm using a real-world road network of Marikina City, Metro Manila, Philippines. The road network consists of 4,892 edges and 3,309 nodes. A total of 20,000 routes were generated for analysis: 10,000 using the baseline shortest path routing (SPR) and 10,000 using the proposed flood-aware routing (FAR). In each trial, a depot node and a required edge to traverse were randomly selected. The randomness ensures that route planning covers a wide spatial and structural range of possible trips within the network. To assess whether our route pairings sufficiently represent diverse spatial configurations, we evaluated the entropy of the depot and edge distributions. Both depot and required edge selections exhibit high normalized entropy ($H_{\text{norm}} \approx 0.976$), indicating broad diversity and a lack of concentration or sampling bias. This confirms that our sampling strategy achieves near-uniform coverage across the network. We evaluate two routing strategies: (1) Shortest Path Routing (SPR), which minimizes distance without considering flooding, and (2) Flood-Aware Routing (FAR), which integrates real-time flood depth and exposure risk via a time-dependent cost function. FAR employs a weighted combination of travel time and risk, parameterized by the risk–time trade-off coefficient $\lambda = 0.1$, slowdown factor $\alpha = 0.3$, and flood threshold $\theta = 0.1$ meters. For each route, we assess: (1) *Travel Time*, the total time adjusted for speed penalties due to flooding; (2) *Flood Risk Score*, the cumulative weighted risk across all segments; (3) *Cumulative Flood Depth*, the aggregate water depth encountered; and (4) *Flooded Segment Count*, the number of segments where depth exceeds θ. We conduct sensitivity analyses across parameter settings and analyze spatial usage patterns by computing edge usage frequency over multiple trials. Spatial clustering of frequently used segments is quantified using Moran's I statistic to identify concentration patterns in routing behavior.

IV. RESULTS AND DISCUSSIONS

The 20,000 routes are generated and analyzed by plotting all the routes and overlaying them per algorithm, shown in Figure 2. The spatial autocorrelation analysis reveals whether frequently used road segments are spatially clustered, dispersed, or randomly distributed. The top row of Figure 2 displays Moran scatterplots for each scenario. The x-axis represents the standardized (z-score) edge usage count, while the y-axis denotes the spatial lag of the z-score. A positive slope indicates spatial clustering. All three distributions exhibit statistically significant spatial autocorrelation at $p < 0.001$, suggesting that frequently used road segments are spatially correlated rather than scattered randomly across the network. The center plot reveals a relatively high Moran's I value of 0.363, indicating strong spatial clustering of edge usage. This implies that FAR tends to prefer specific corridors or sub-networks as optimal evacuation paths under flood risk. These high-usage corridors are likely flood-resilient, topographically elevated, or possess multiple alternative paths, like essential connections between two highly separated network communities, making them robust to dynamic flooding scenarios. The rightmost plot reports a Moran's I value of 0.336, slightly lower than FAR. Although spatial clustering remains prominent, the pattern reflects the static and deterministic nature of shortest path calculations. Since SPR does not account for flood dynamics,

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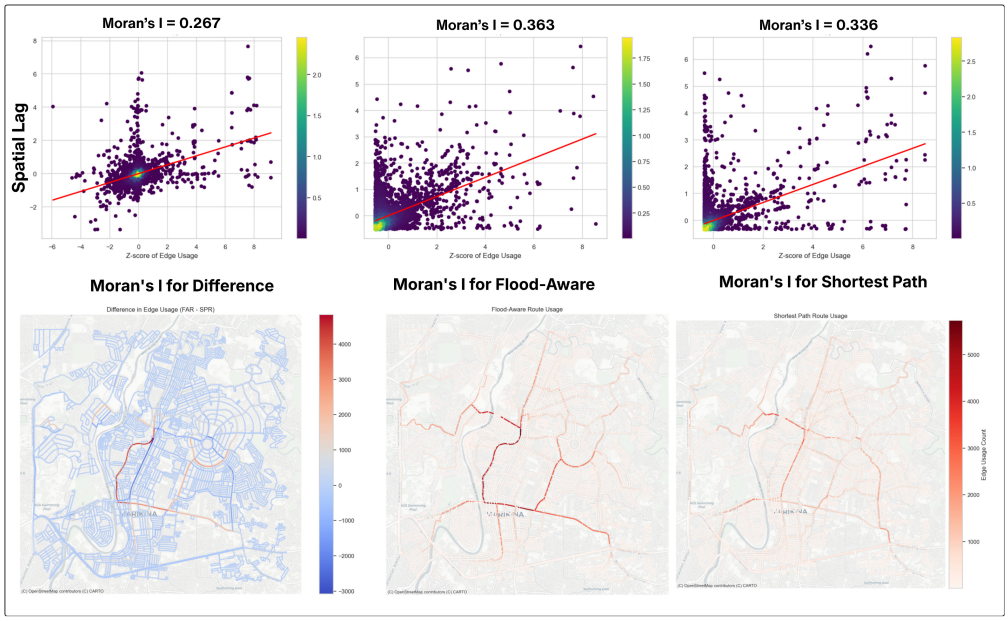


Fig. 2. Moran's I Spatial Autocorrelation and the Corresponding Route Segment Usage.

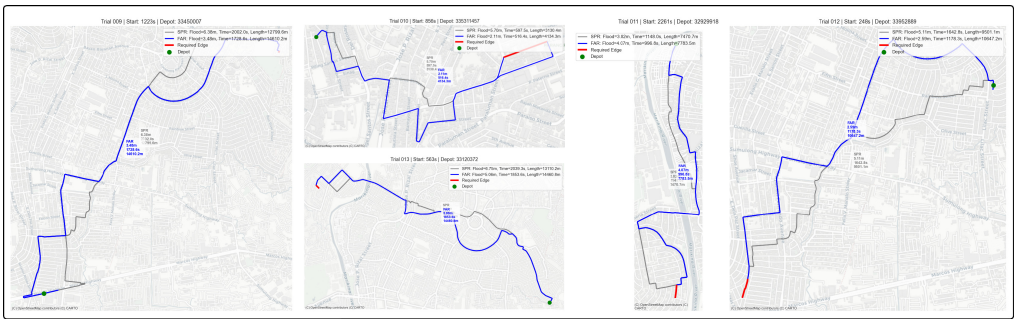


Fig. 3. Route Set 1.

it consistently favors geometrically shortest routes which may not adapt over time, explaining the relatively lower clustering compared to FAR. The leftmost plot shows a Moran's I of 0.267 for the difference in edge usage counts between FAR and SPR. This captures the extent to which route selection shifts between the two algorithms in a spatially structured manner. The lower but still significant autocorrelation implies that changes in edge usage are not randomly distributed. Rather, these differences concentrate in certain areas where FAR systematically avoids (or prefers) segments due to flood hazards, particularly in flood-prone zones that SPR naively includes. The bottom row of Figure 2 visualizes these patterns geographically. In the FAR usage map, specific arterial roads exhibit elevated usage density, highlighted in darker red tones. In contrast, the SPR map shows a wider but more uniform spread of traffic with slightly different spatial clusters. The difference map emphasizes the exact segments where FAR diverges most from SPR—highlighting segments that FAR avoids (blue, negative values) versus those it overuses relative to SPR (red, positive values). These deviations reveal FAR's ability to reroute away from flooded segments, reinforcing its adaptivity. The upper subplot of Figure 5 displays the 15 most frequently used edges in the entire route dataset, based on total usage count across both routing strategies. From here, some edges are heavily shared by both strategies, suggesting they serve as core arterial links within the network. However, a notable proportion of edges exhibit higher usage in FAR, indicating a rerouting behavior influenced by flood dynamics. In particular, FAR demonstrates preference for alternative links that would not typically be prioritized under the distance-based baseline, implying adaptation to local topographic or hydrological conditions. In terms of most utilized highway types, `tertiary` and `residential` roads dominate, followed by `primary` and `secondary` roads. This reflects

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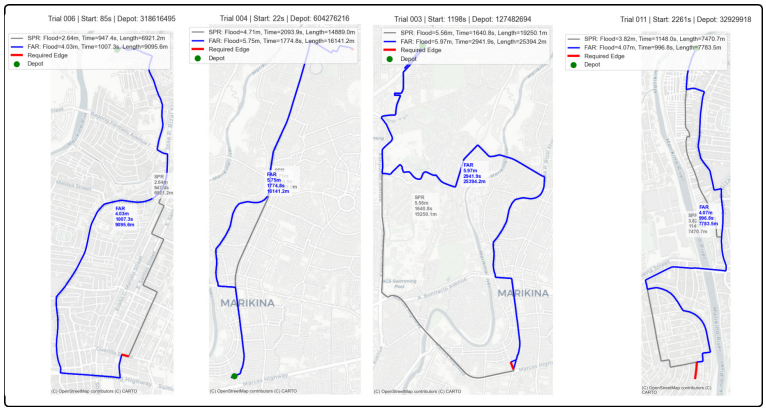


Fig. 4. Route Set 2.

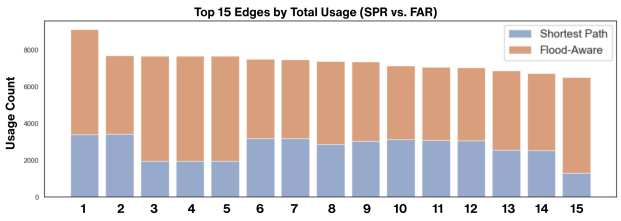


Fig. 5. Top Edges by Total Usage: SPR vs. FAR.

the urban layout of the study area where local and connector roads form the backbone of intra-city mobility. Interestingly, FAR's reliance on `residential` and `tertiary` roads likely reflects a strategic avoidance of known flood-prone thoroughfares. Minor usage is observed for categories like `living_street`, `busway` indicating that auxiliary routes and connectors play a limited role in both strategies. This findings has implications for flood-resilient infrastructure planning, as it highlights which street types and links are most critical during emergencies. The results of the sensitivity analysis is shown in Figure 6. As the flood threshold increases from 0 to 0.3 meters, the model becomes more tolerant of flooded segments. This leads to a steep decline in flood risk, travel time, and flooded segment count. However, cumulative flood depth remains relatively stable across thresholds, slightly increasing with higher θ due to inclusion of mildly flooded paths. This suggests that overly conservative thresholding results in excessive rerouting with minimal gain in risk reduction. The λ parameter controls the tradeoff between travel time and flood risk in the cost function. As λ increases from 0 to 1.0, the flood-aware routing strategy prioritizes travel time minimization leading to a decrease in both risk and travel time. The α parameter penalizes speed reductions in flooded segments. As α increases, travel time and flood risk both decrease, indicating more aggressive avoidance of high-depth segments. However, this shift also causes an increase in the number of flooded segments traversed and the cumulative flood level. The implication is that while high α values enable faster routing under flooded conditions, they may encourage shorter but more flood-prone paths, accepting shallow flooding more frequently. The risk constraint imposes a hard upper bound on the total allowable flood risk for any route. When `risk_limit` increases from 0.1 to 2.0, all four metrics (risk, time, flooded segments, cumulative flood) stabilize after the initial jump from 0.1 to 0.5. This suggests that beyond a modest constraint level, the optimization operates unconstrained in practice. Only overly strict limits materially affect route selection and force detours. Therefore, `risk_limit` serves as a soft safety override but does not significantly influence routing outcomes under normal conditions. Figures 3 and 4 visualizes several route pairs under varying flood conditions and temporal start times. In some cases, FAR succeeds in significantly reducing cumulative flood exposure compared to SPR (e.g., Trial 010: 2.11m vs 5.70m), albeit with slightly longer travel times. In other cases, the benefits are more nuanced as FAR avoids major flood zones early but may end up in flood-heavy areas later on (e.g., Trial 003). This behavior can be attributed to the local and greedy nature of the routing algorithm. The engine selects the next edge based on minimizing a dynamic cost function, without full foresight of downstream flood conditions. Consequently, it lacks the capacity to recognize that a short detour early in the route may prevent worse flooding in the future. This short-sightedness highlights the inherent limitations of local decision-making in dynamic flood environments. Urban networks, especially under extreme conditions, often present non-linear tradeoffs between risk and time. As shown in multiple examples, avoiding a moderately flooded edge may result in a path that ultimately encounters more severe flooding or incurs excessive delay. Overall, the comparative evaluation across multiple routing configurations, spatial usage patterns, and parameter sensitivities demonstrates that while flood-aware routing can substantially reduce inundation exposure, it requires careful tuning and deeper understanding of the evolving risk landscape to consistently outperform baseline strategies. Building on these findings, the next section presents the concluding insights

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of the results and offers methodological recommendations for enhancing flood-adaptive routing systems.

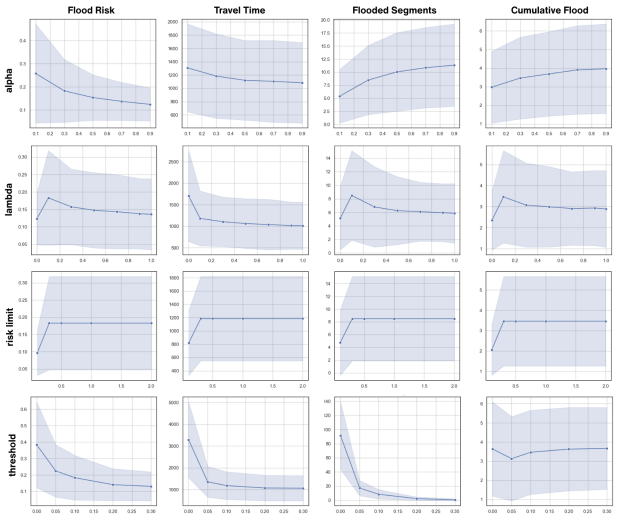


Fig. 6. Parameter Sensitivity

V. Conclusions and Recommendations

This work presented a dynamic flood-aware routing model for urban evacuation using an Arc Routing Problem (ARP) formulation over a real-world network in Marikina, Philippines. Experimental results show that the flood-aware routing strategy (FAR) reduces flood exposure and number of flooded segments compared to the shortest-path baseline (SPR), albeit with slightly increased travel time. From a spatial perspective, through Moran's I, non-random clustering is revealed in route selection and segment usage, indicating potential bottlenecks or vulnerable zones that warrant further planning attention. Parameter sensitivity analyses showed that low flood thresholds (θ) and high flood penalties (α) result in risk-averse behavior but longer paths. Conversely, adjusting λ allowed control over the flood–time tradeoff. While FAR is locally flood-averse, it may fail to avoid worse conditions downstream due to greedy path selection, as detailed in the previous section. Route performance is highly sensitive to spatial heterogeneity in flood conditions and network topology. The observed variation in routing performance can be attributed to multiple interacting factors including localized flood intensity, heterogeneous road connectivity, and the structural constraints of the road graph. This highlights the importance of designing evacuation plans that are responsive to spatial heterogeneity in both hazard distribution and infrastructure layout. While the current model demonstrates an adaptable framework, further advancements are needed to address long-term risk planning under uncertainty and the need for routing algorithms that are not purely greedy, but rather hybridized strategies. Future extensions of this work may consider multi-objective routing algorithms, multi-pass strategies, multi-criteria Dijkstra variants, or global optimization techniques to better balance short-term sacrifices against long-term gains. This includes integrating predictive flood forecasting, global situational awareness, and decision-making under incomplete or evolving information to enhance the robustness and foresight of evacuation strategies.

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